NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

Requests shall be referred to Headquarters, U.S. Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, MD 21005-5055, usarmy.APG.atec.mbx.atec-standards@mail.mil

Custodians:
Army – TE

Preparing activity:
ATC

AREA ENVR

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

REFERENCE FILE

TECP 700-700
Materiel Test Procedure 3-2-801*
Aberdeen Proving Ground

27 October 1965

U.S. ARMY TEST AND EVALUATION COMMAND
COMMON ENGINEERING TEST PROCEDURE

MEASUREMENT OF INTERNAL DIAMETERS OF CANNON

1. OBJECTIVE

The objective of this document is to instruct personnel in the techniques of measuring the internal diameters of cannon, to familiarize them with the two basic types of gages available for these measurements and to describe the methods of measuring the internal diameters of cannon.

2. BACKGROUND

The condition, i.e. remaining life, of the bore and chamber of cannon is indicated in part by the diameters at various locations. These measurements are applicable in the testing of weapons and in the evaluation of bore wear or damage. A comparison of the gages available, and their method of operation is discussed in Appendix A.

3. REQUIRED EQUIPMENT

a. Pullover Gage
b. Small Bore Stargage
c. Lever Type Stargage
d. Auxiliary Stargage Equipment
    1) Staves as required
    2) Exterior Rests
    3) Size Rings
    4) Guide Plates
    5) Measuring Points
    6) Bore Guides
    7) Bore Supports
    8) Circular Discs to aid in centering
e. Platform (when axis of bore is more than 54 inches from ground) equal in length to the cannon.

4. REFERENCES

A. TM9-1000-202-35 Evaluation of Cannon Tubes, Change 1: 27 March '64
B. TM9-4933-200-35 Operation and Field and Depot Maintenance; Pullover Gages, Borescopes, M1 and M2, and Pressure Gage M3: 12 August '58

*Supersedes Ordnance Proof Manual 80-41

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

5. SCOPE

5.1 SUMMARY

This MTP pamphlet describes the pullover gage and stargage mechanisms and discusses their use and application. Included are the operating principles, selection of equipment, preparation for gaging, and the gaging procedure.

5.2 LIMITATIONS

Cannon chamber slopes with a diametral taper of 0.100 inch per inch or greater will not be gaged.

6. PROCEDURE

6.1 PREPARATION FOR TEST

6.1.1 Stargaging

To insure accuracy of final results, preparation for stargaging must be meticulous. Preparation of the cannon bore and the sire of the operation are equally as important as proper assembly and operation of the gaging equipment. Stargaging shall not be performed in the open during inclement weather.

6.1.1.1 Preparation of Cannon

Cannon normally are prepared so that gaging can be conducted from the breech end. With cannon of 12-inch diameter or larger, in tank armament, or on weapons with characteristics that prohibit gaging from the breech, measurements shall be taken from the muzzle. The bore and chamber of the cannon shall be cleaned thoroughly and dried.

6.1.1.2 Preparation of Site

All cannon to be stargaged will be located with ample working space at either end. The end from which the gaging is to be conducted must be clear from a distance at least equal to the length of the weapon. The opposite end of the bore shall be clear for not less than 2 feet. When the axis of the bore is more than 54 inches from the ground, a sturdy platform equal in length to the cannon, will be erected at the end from which the operation is to be conducted.

6.1.1.3 Preparation of the Stargage

Select and assemble a stargage of the proper size (see Appendix A). The measuring head shall be assembled to the necessary staves, bore rests, measuring points, and actuating handle. The gage will then be inserted in the bore of the weapon. Exterior rests will be placed to maintain the exterior sections of the gage coaxial with the bore (see Fig. 1). The instrument is set to the nominal diameter of the weapon by employing a size ring (see Fig. 2). With the zero adjustment locked

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

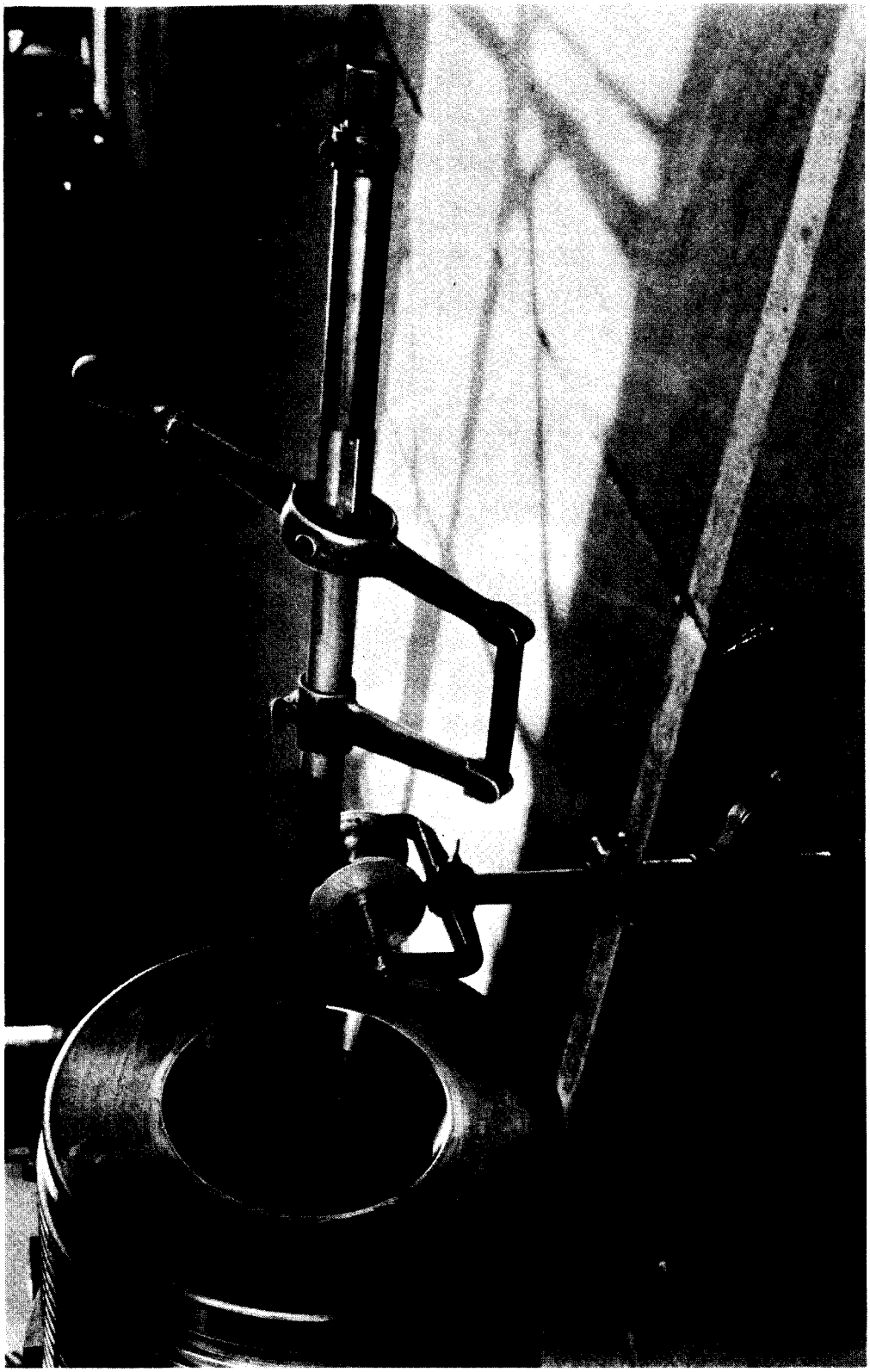


Figure 1. Exterior Rest in Position for Gaging Operation

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

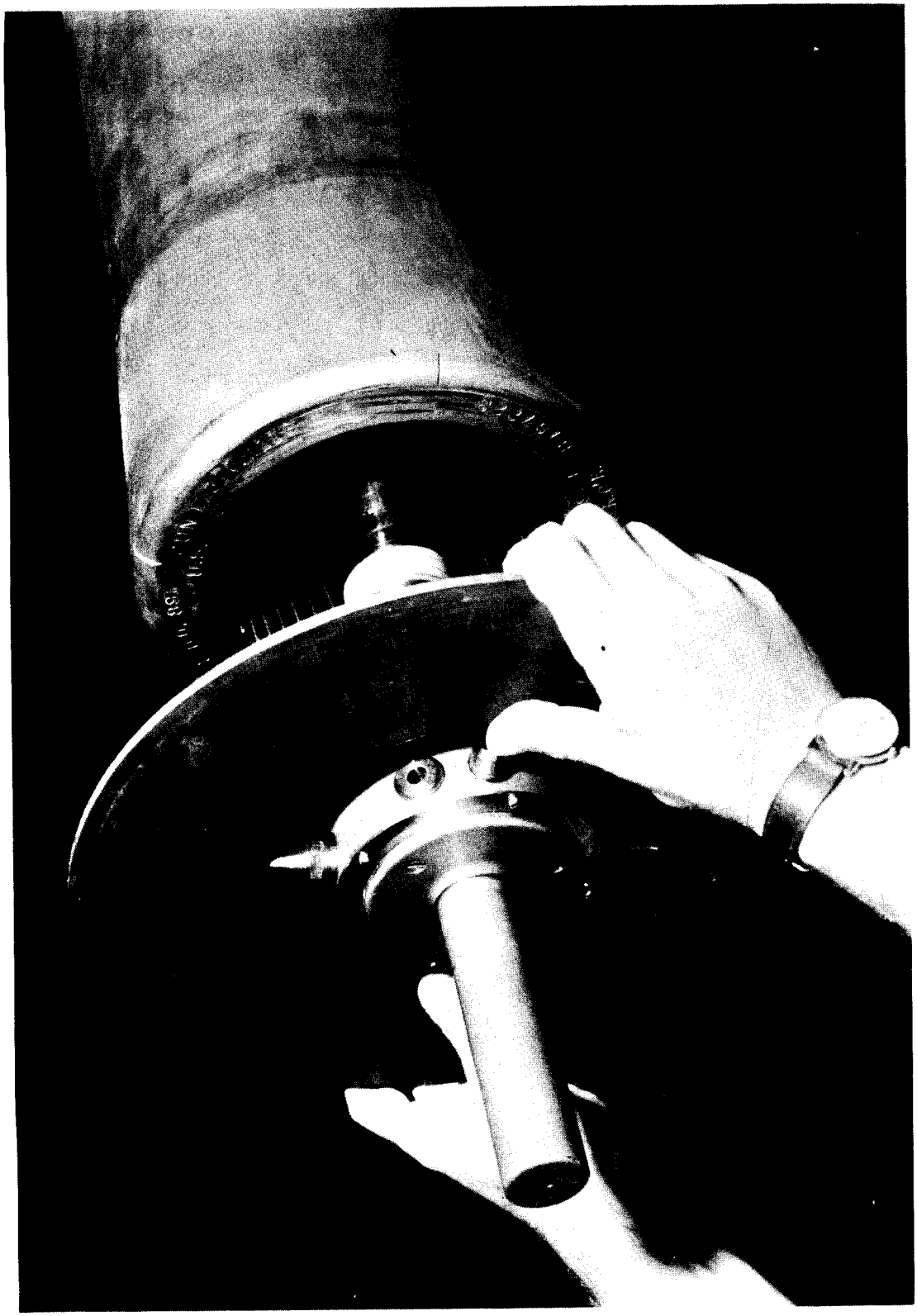


Figure 2. Setting Stargage with Size Ring

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

and a guide plate affixed to the measuring head to maintain the measuring points upon the desired land or groove, the gage is ready for operation.

6.2 TEST CONDUCT

6.2.1 The Stargage

The procedure for gaging gun bores shall be followed carefully. Insofar as conditions permit, the procedure must be identical for every run in the bore and chamber.

6.2.1.1 Stargaging the Cannon Bore

All stargage measurements made in a cannon bore will be relative to the "vertical land", which is defined as the land located at the 6 o'clock position at the origin of rifling when the cannon is in a normal position. The adjacent groove in a clockwise direction, as one faces the muzzle, is designated the "vertical groove". The vertical land and groove location at the muzzle shall be computed from the drawing for the individual model cannon or by following the vertical groove from the origin of rifling to the muzzle with the stargage.

One set of measurements shall be made on the vertical and horizontal lands and another in the vertical and horizontal grooves throughout the length of the bore. When a three-point head is employed, a Y-up and a Y-down set of measurements shall be made on both lands and grooves. For the Y-up position, the vertical land forms the base of the Y; for the Y-down position, the land diametrically opposite the vertical land forms the base. After determining the correct vertical or Y-up position, place the measuring head in the bore of the cannon (see Fig. 3) and secure the appropriate guide plate to the head to maintain the measuring points on the desired land. The intervals at which measurements are made in the bore will vary from 2 to 5 inches, depending upon the nominal diameter and length of the bore. Measurements in the vicinity of the commencement of the rifling, where wear and erosion are greatest, will be made at smaller intervals.

The distance at which measurements shall be made are standardized for each type and model of gun bore measured. Normally, these measurements are made from 0.10 inch forward of the commencement of rifling to 0.10 inch from the muzzle extremities of the rifling. These distances shall be computed from the drawing of the cannon tube. In operation, position the gage at the appropriate gaging stations, obtain the diameter at that station, and record the obtained measurement on the field form. After passage of the gage through the bore, with the measuring head revolving with the twist or rifling, advance the gage until the measuring head is at the muzzle. Succeeding runs will be made until the required measurements have been completed.

In badly worn or eroded cannon, it is often necessary to obtain precise measurements in the damaged area. For these checks, lock the measuring head with respect to the staves so that turning the staves

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

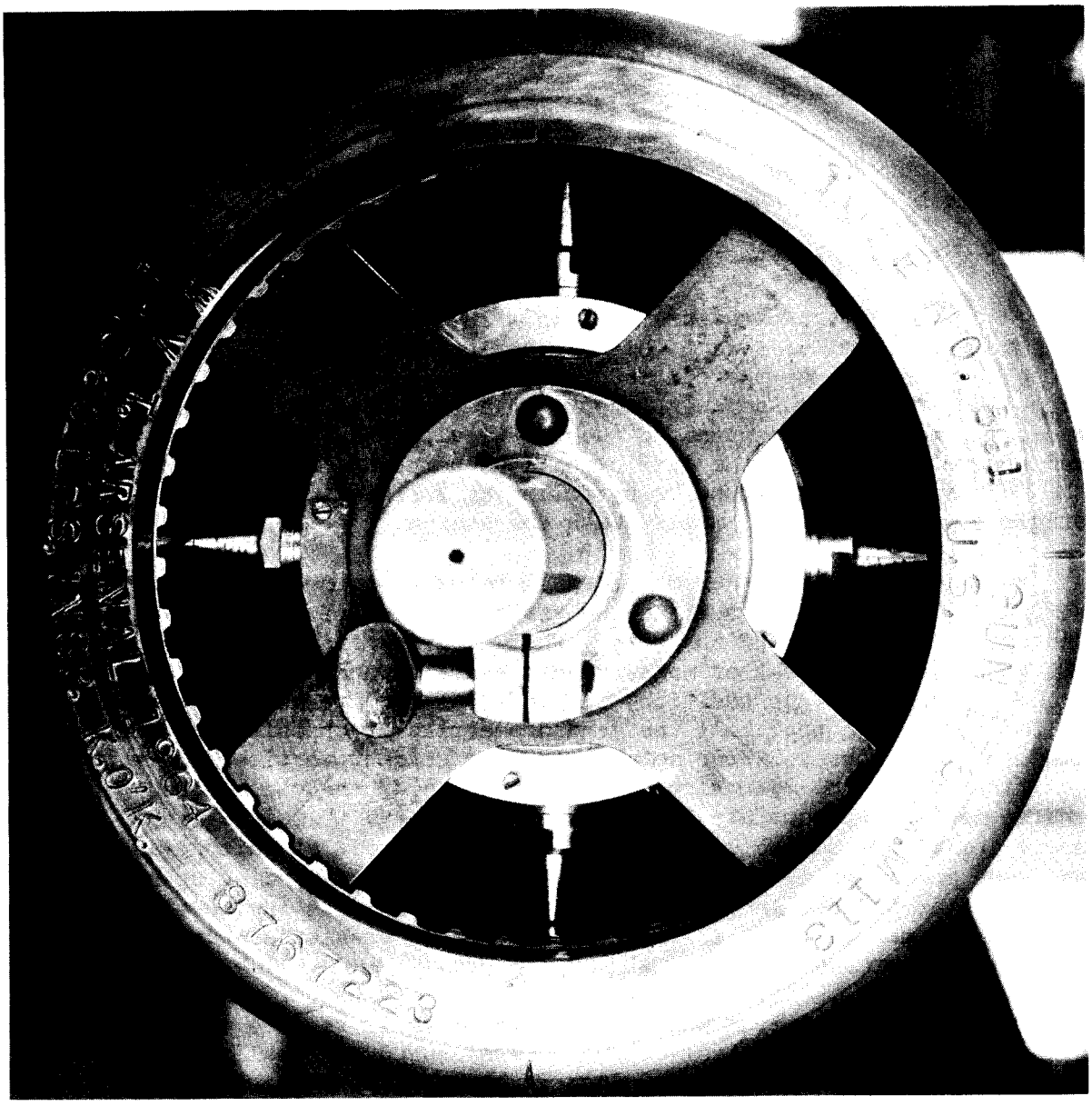


Figure 3. Measuring Head Emplaced in Cannon at Muzzle

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

will cause the head and points to rotate through the same angle. The results obtained depend on the operator's sense of touch and the centering of the measuring head in the cannon bore. To aid in the centering, circular discs of various sizes may be fabricated and attached to the measuring head. When rifling has become obliterated, the minimum diameter at a given point will be considered as the land diameter and the maximum considered as the groove diameter. In such cases, a notation will be made on the gaging record stating that measurements taken in such areas represent a maximum and a minimum diameter.

6.2.1.2 Gaging the Chamber

When measurement of the rifled bore has been completed, the stargage shall be assembled to gage the chamber. The chamber will normally be gaged with a three-point head because of the taper present in most gun chambers. The procedure for chamber measurements follows that for the gun bore, chamber slopes with a diametral taper of 0.100 inch per inch or greater will not be gaged because it is difficult to obtain accurate measurements on sharp slopes. Gaging stations in the chamber are usually located at smaller intervals than in the rifled bore. Stations are determined by consulting the chamber drawing for the individual model cannon, and extend throughout the measurable areas of 0.10 inch of their extremities. The basic chamber diameter for each gaging station shall be noted on the stargage report, and the variation from the basic dimension at each location will be determined by the gaging operation.

6.2.2 The Pullover Gage

The cannon and site shall be prepared as indicated under paragraphs 6.1.1.1 and 6.1.1.2.

The pullover gage functions somewhat as a telescoping inside micrometer. To operate the following functions must be accomplished:

a. Extend the slide (see Fig. 4) beyond the body of the gage until the length of the gage exceeds the expected diameter of the bore.

b. Tilt the gage forward and insert it into the bore with the exposed end of the slide up.

c. "Pull Over" the gage (see Fig. 5) and force the slide into the body.

d. Measure the gage length, this is equal to the minimum diameter of the bore at the location being measured. The variation of the diameter from the basic value is read from the vernier scale on the gage.

6.3 TEST DATA

6.3.1 Stargaging

6.3.1.1 Cannon Bore

Record the following for the entire length of the bore, at specified intervals (two to five inches) to within .001 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

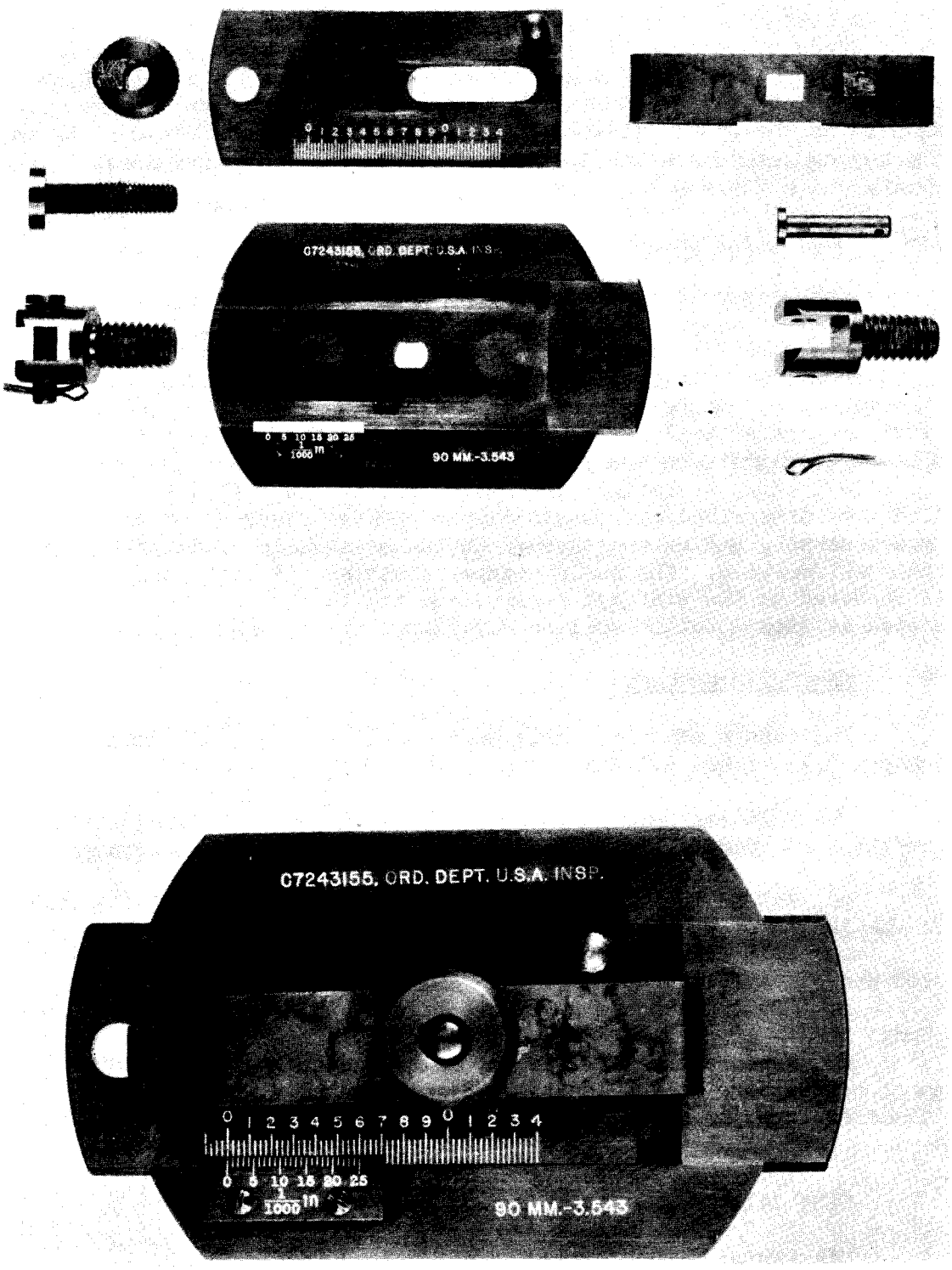


Figure 4. The Pullover Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

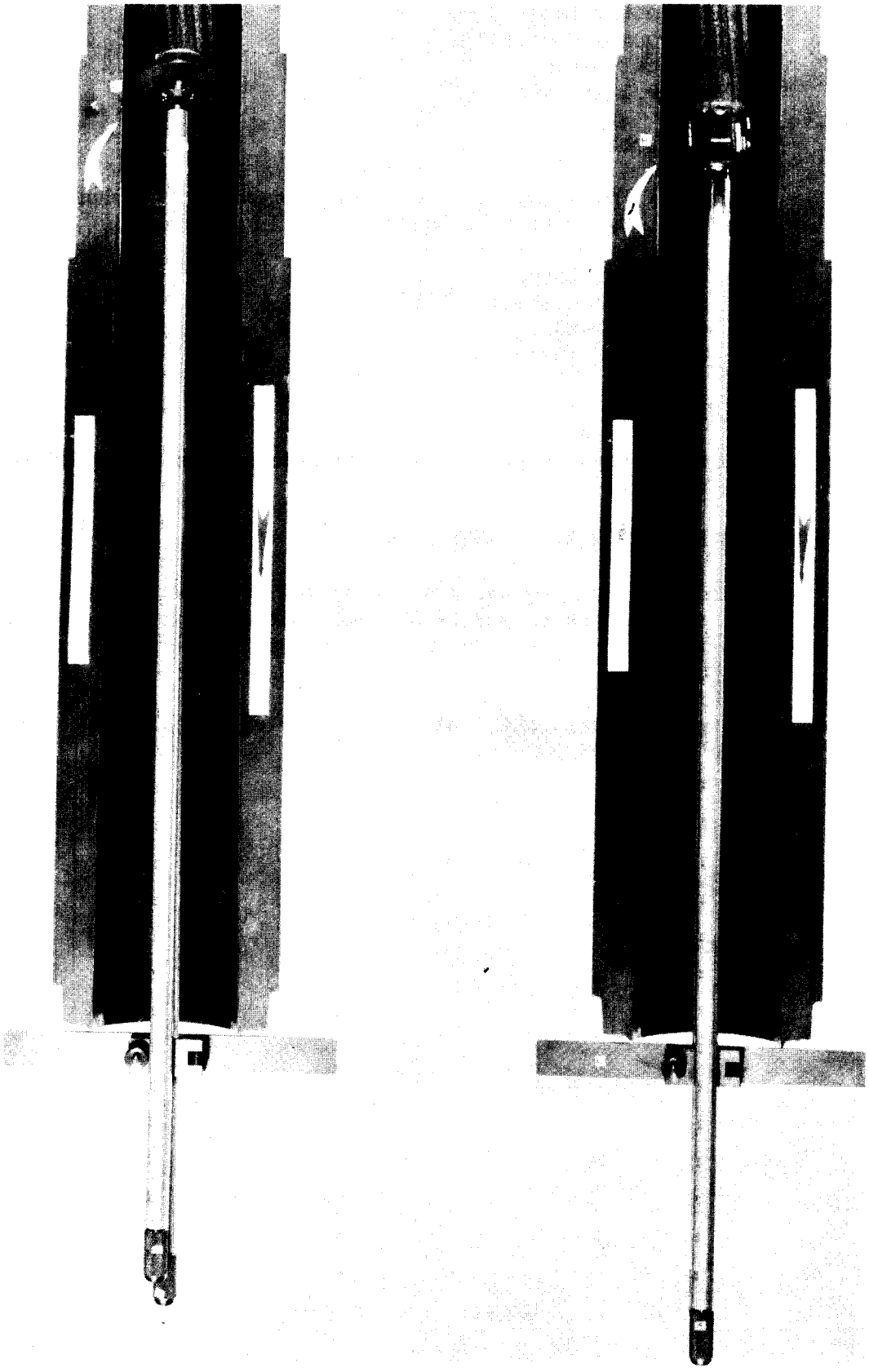


Figure 5. Assembly and Operation of the Pullover Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

a. Horizontal Lands
b. Horizontal Grooves
c. Vertical Lands
d. Vertical Grooves

6.3.1.2 Cannon Chamber

Record the following for the entire length of the chamber at specified intervals (two inches or less) to within .001 inch.

a. Horizontal Lands
b. Horizontal Grooves
c. Vertical Lands
d. Vertical Grooves

6.3.2 Pullover Gage

Record the gage length (minimum diameter) at specified intervals to within .001 inch.

6.4 DATA REDUCTION AND PRESENTATION

Agencies conducting the stargaging and pullover gage measurements shall maintain a permanent record of all data obtained and insure that appropriate entries are made in the weapon record log (Weapon Record Data, DA Form 2408-4).

The recorded data shall be interpreted as stipulated in TM 9-1000-202-35 and TM 9-4933-200-35.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

APPENDIX A

OPERATING PRINCIPLES OF STAR AND PULLOVER GAGES

Two types of gages are presently available for the measurement of diameters of cannon bore: the pullover gage, used principally to measure the diameter at a single location within the bore; and the star-gage, used for measurements within the chamber and throughout the length of the bore. These devices and the techniques used to measure these inside diameters are considered here.

In general, the stargaging method is usually used for measuring the internal diameters of cannon tubes. However, the pullover gage has certain advantages that make it a useful supplement to the stargage and a primary instrument for evaluating serviceability. Since the pullover gage is more rugged than the stargage, it is used extensively as a field gage for quick determination of cannon tube wear. TM's 9-1000-202-35 and 9-4933-200-35 describe the pullover gage, its application, and the interpretation of results in detail.

A direct comparison of the performance of the two gage types is difficult. In a new tube with unbroken land surfaces, the gages, in effect, are measuring a smooth walled cylinder, and nearly identical values will result when measurements are made at the same position within the bore. The differences between readings made with both gages under these circumstances generally result in a standard deviation of less than 0.002 inch. Thus, the relative accuracies of the gages are comparable. On the other hand, as the bore surface becomes eroded, the relative accuracies of the gages are not pertinent because of the difference in the characteristic of the gages. The stargage's probe-like points contact a limited position of the circumference of the bore section being measured. However, the bearing edge of the pullover gage is wide by comparison and has a radius only slightly less than that of the bore. This subjects the pullover gage to the influence of high spots lying on arcs many times longer than those contacted by the stargage points. In light of this, the user is cautioned to consider the significance of the gaging values rather than to be concerned with the relative accuracies of the gaging mechanisms.

The precision of measurements, or reproducibility of results, realized with the pullover gage when used by experienced operators is represented by a standard deviation within 0.002 inch. The precision of measurements realized with the stargage is comparable.

THE STARGAGE

The stargage is a precise measuring instrument for determining the internal diameters of cannon. This gage also represents an extension of the principle of the inside micrometer.

Many instruments have been developed for measuring the interior diameter of objects of circular bore. Those that have proven most suitable for cannon are the stargages, including the lever type, micrometer type, spring-actuated dial-indicator type, and the small bore gage designated for small arms. Since the micrometer gage has been supplanted by the lever

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

gage for general Army application, and the dial gage is suitable only to new or slightly worn cannon and is therefore limited in use, only the lever type stargage and the small bore gages are discussed herein.

For cannon of specific diameters, select stargages of the appropriate nominal size from the table under Lever Stargage. Select the instrument and accessories which will give adequate ranges to encompass any diametral increase anticipated. Allowance must be made for sufficient negative range to permit disengaging the measuring points.

The Lever Stargage

The lever stargage is available in four sizes (see Fig. A-1) each designed for specified diametral ranges as indicated in the table which follows:

Lever Stargage Ranges

Nominal Size (in.)	Minimum Diameter Measured (in.)	Cannon Bore for Which Used
0.75	0.75	20- to 30-mm
1.35	1.35	37- to 75-mm
2.00	2.00	75- to 155-mm
5.75	6.00	155-mm to 16-inch

The lever stargage is composed of three major assemblies: the measuring head, the staff, and the operating handle. The measuring head, operating handle, and various other components of the 5.75-inch lever stargage are shown in Figure A-2.

Measuring Head

Two head assemblies are provided with each lever gage: one with three movable measuring points, the other with four points, two of which are movable. Selection of the head assembly must be predicated on several factors. The number of lands and grooves in the cannon bore must be a multiple of the number of measuring points. The type of gaging operation contemplated must be considered. Routine gaging of acceptance weapons is frequently conducted with the three-point head. When advanced erosion is encountered or when the cross section of the bore varies from a true circle, the extreme dimensions of out-or-round bores or eroded area are obtained more accurately with a four-point head. With the three-point head, the measurements obtained under these conditions represent an average of three diameters.

Measuring points, bore guides, size rings, extra staves, and bore supports are provided for the nominal size of each standard cannon bore. When variations from standard diameters are encountered, adaptations or modifications of standard equipment can usually be made with little

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

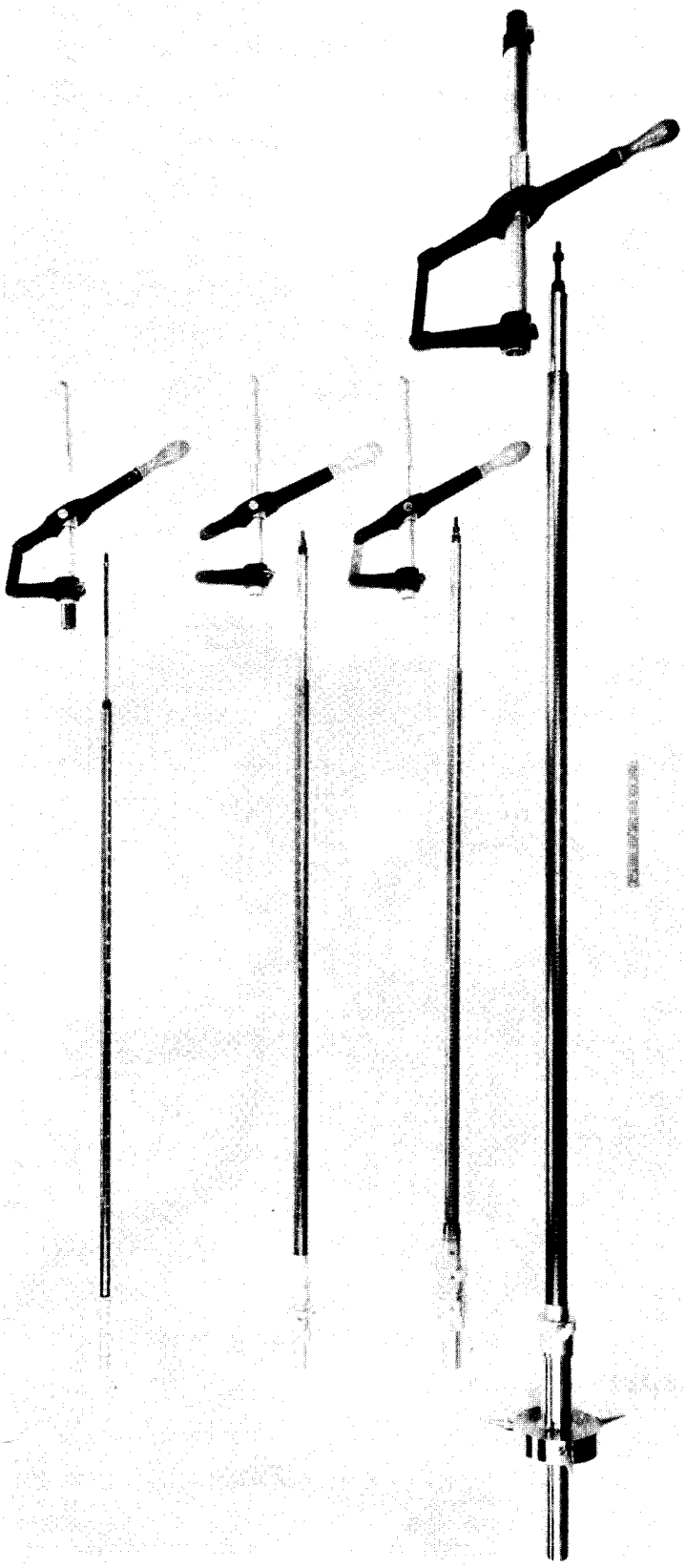


Figure A-1. Lever Type Stargages

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

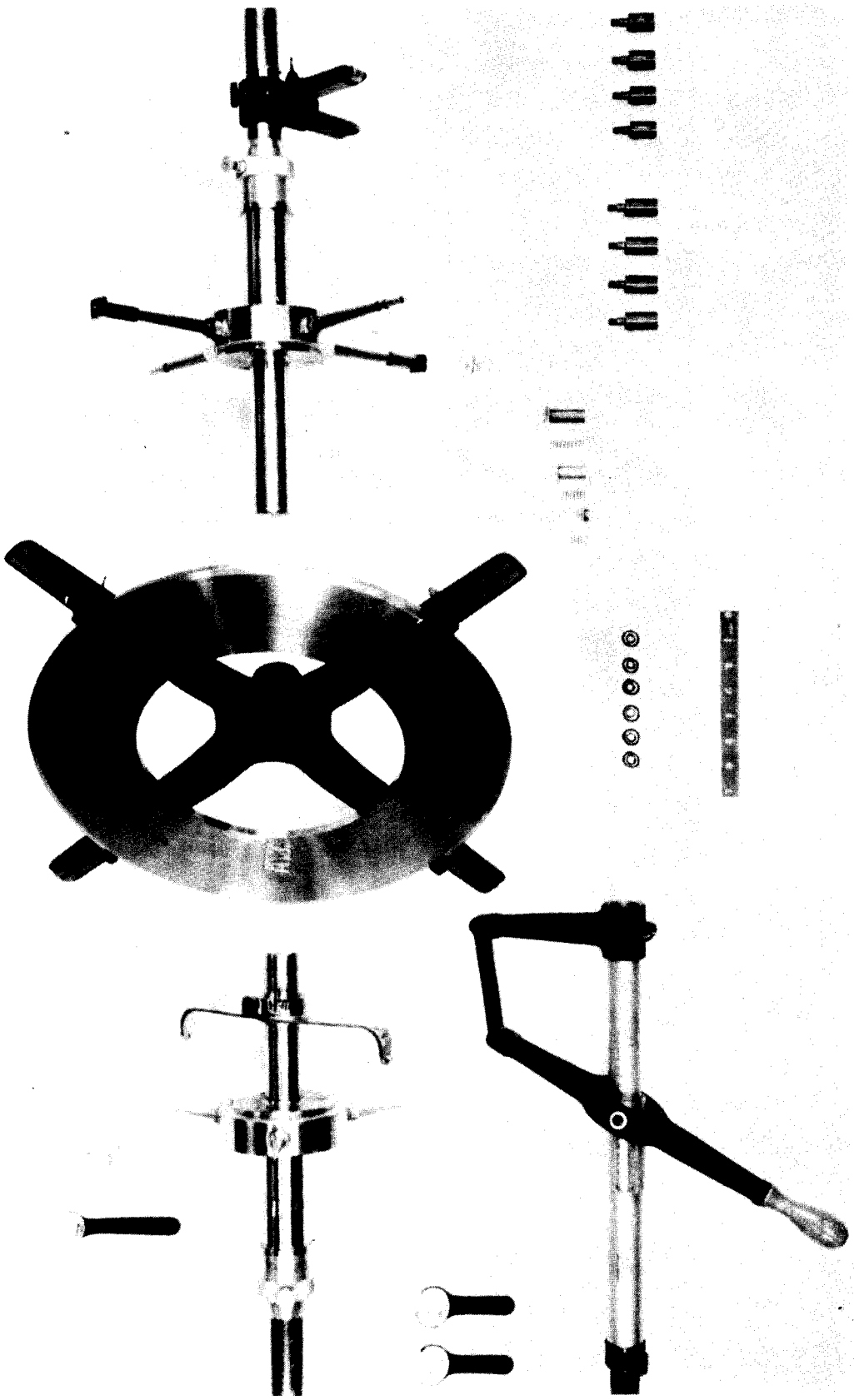


Figure A-2. 5.75-Inch Lever Stargage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

difficulty.

The heart of the measuring head is a finely tapered cone or wedge attached to an actuating rod. Movement of the rod imparts motion to the cone or wedge, which, in turn, actuates a set of movable bushings to which measuring points are attached. As the bushings are forced out at angles of 90° to the axis of the cone or wedge, the measuring points extend outward and make contact with the surface to be measured. Upon rearward motion of the rod, the cone or wedge retracts, permitting the measuring points to disengage. The head may either revolve independently of the staff or may be locked in position.

The Staff

Three functions are served by the staff. The outer cylinder serves as a rigid connecting member between the measuring head and the operating handle. The inner rod is supported coaxially with the outer cylinder in such a manner as to permit longitudinal movement of the inner rod and forms a connecting link which transmits motion from the operating handle to the measuring head. The outer cylinder is graduated to aid in determination of the distance to the point of measurement. Any number of staves may be joined to meet distance requirements.

The Operating Handle

The operating handle is composed of a rigid inner cylinder attached directly to the outer cylinder of the staff, a movable outer cylinder which attaches to the inner rod of the staff, and an actuating lever. Any movement of the lever produces a diametral change at the measuring points. Attached to the operating handle is a calibrated vernier scale graduated to 0.001 inch.

The Small Bore Stargage

The small bore stargage (see Fig. A-3) also employs the principle of the tapered cone actuated by a movable inner rod. The operating handle, however, is an extension of the inner rod and is operated by longitudinal motion within the rigid outer cylinder. Readings are made from calibrations on the inner rod where they coincide with an index on the rearward end of the outer cylinder. Unlike the lever-type gage, the small bore gage does not have a revolving measuring head and, because of its limited range, different size gages are required for different calibers.

THE PULLOVER GAGE

The pullover gage is used primarily to evaluate the serviceability of cannon tubes. Studies of tube wear show that single measurements slightly forward of the commencement of rifling may be correlated with serviceability and remaining life. For condemnation of most tubes, criteria of maximum wear have been established in terms of pullover gagings (TM 9-1000-202-35).

Since it may be inserted or withdrawn through openings much

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.



Figure A-3. Caliber .30 Small Bore Stargage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

smaller than the diameter to be measured, the pullover gage has an added application in that it may be used to supplement stargage observations. For example, when a bulge within a tube bore exceeds the range of a stargage, a pullover gage may furnish the desired measurement.

In badly worn tubes where the rifled surface beyond the commencement of rifling is very uneven, use of the pullover gage is often preferred. Since its results are unaffected by pits, scoring marks, and such, the pullover gage will provide data representative of the effective tube diameter. Here again, the pullover gage may supplement the stargage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:49Z
Check the source to verify that this is the current version before use.